

Nagarajan Shunmugam

Veterinary medicine · CRISPR and NGS machine learning · Autonomous systems and applied AI
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PROFILE

My path has not been a straight line. Electrical and electronics engineering in Chennai, biomedical instrumentation at IIT Madras and IIT Delhi, a dual MSc in Autonomous Systems (Robotics and AI) between TU Berlin and KTH Stockholm on a full EIT Digital scholarship, autonomous trucks at Scania, clinical imaging AI in Berlin, and two years of machine learning for the world's largest veterinary community. Since November 2025 I have been at the **Ludwik Hirszfeld Institute of Immunology and Experimental Therapy** (Polish Academy of Sciences) building CNN and Transformer models for CRISPR guide design and off-target prediction, and doing the bench work that generates the data. In December 2026 I qualify as a veterinarian.

The three strands are one interest, not three careers. Engineering taught me to build systems that decide under uncertainty; genomics gave me a domain where the decisions are consequential and the data is unforgiving; veterinary medicine keeps me honest about the animal at the end of the pipeline. I have co-founded three companies, published, taught for six years, and shipped tools people actually run.

SELECTED EXPERIENCE

CRISPR & NGS Data Researcher · [Ludwik Hirszfeld Institute, Polish Academy of Sciences](#) 2025 – Present | Wrocław

- Machine-learning pipelines for CRISPR off-target prediction and guide design on genomic, NGS and multi-omics data — k-mer, one-hot and scalar encodings into CNN and Transformer models, with reproducible cross-validation and quality metrics.
- Multi-agent AI systems that design, schedule and validate wet-lab experimental protocols.
- Bench work alongside the computation: sequencing library preparation, validation assays, PCR template preparation, gel verification and cleanup, mRNA synthesis and purification.
- Deployed two research tools — G-Cat, for CRISPR, base-editing and prime-editing guide design, and a companion RNA-CRISPR tool. Two manuscripts in preparation.

Co-Founder & CTO · [MilkAI LLC](#) 2025 – Present | California

- Product studio turning ideas into production web, mobile and AI-first applications, with design-to-deploy CI/CD. Dozens of launches; ownership of architecture, delivery and roadmap.

Artificial Intelligence Engineer · [Veterinary Information Network \(VIN\)](#) 2023 – 2025 | Remote, US

- Machine learning for the world's largest online veterinary community, including a domain-specialised clinical language model with engineered hallucination reduction for practitioner-facing question answering.

Co-Founder · [Zaide.ai](#) 2023 – 2024 | Remote

- Private, on-premise large language models for healthcare and life sciences, built so sensitive clinical and research data never leaves the organisation.

Co-Founder & CTO · [Neurons4Vet.ai](#) 2020 – 2021 | Toronto

- First venture at the intersection of AI and veterinary medicine: classification, segmentation and detection on radiographs, ultrasound and MRI.

Cohort Member · [Entrepreneur First](#) 2021 | Berlin

- Selected for Europe's leading deep-tech talent investor.

MSc Thesis — Visual Perception for Autonomous Trucks · [KTH Royal Institute of Technology & Scania Group](#) 2020 – 2021 | Stockholm

- Multi-task perception on real operational data: ResNet classification, YOLOv3 detection, modified U-Net semantic segmentation with residual skip connections, monocular depth estimation, and reinforcement learning — under fixed latency and memory budgets. Produced an IEEE conference paper.

Machine Learning Engineer · [ai4medicine](#) 2020 | Berlin

- Computer vision for stroke and intracranial haemorrhage detection in patients.

Senior Design Engineer — Driverless Team · [KTH Formula Student](#) 2019 – 2020 | Stockholm

- Autonomous race car perception: object detection, localisation, LiDAR point clouds, NVIDIA Jetson deployment.

Teaching — Assistant Lecturer and Course Mentor · [Coventry University \(Wrocław campus\)](#) | [Coursera](#) 2017 – 2023

- Taught AI, IoT and software engineering at Coventry. Six years mentoring the Neural Networks & Deep Learning course on Coursera, plus alpha-testing unreleased AI courses and mentoring 3D Printing Applications.

Research Assistant — Driverless Car Project and Biomedical Laboratory · [IIT Delhi](#) | [IIT Madras](#) 2017 – 2018 | India

- PID tuning and sensor fusion across GPS, IMU, LiDAR and radar into a complete self-driving pipeline (Delhi). Embedded biomedical instrumentation: SpO2 pulse-oximetry sensing and low-cost clinical screening kiosks (Madras).

EDUCATION

Doctor of Veterinary Medicine (DVM) · [Wrocław University of Environmental and Life Sciences \(UPWr\)](#) 2021 – Dec 2026 (expected)

- Long-cycle degree, 290 ECTS. Immunology, microbiology, parasitology, pharmacology, toxicology, pathomorphology and food hygiene; clinical rotations and necropsy across small animal, farm animal, equine, avian and aquatic species.

MSc, Autonomous Systems (Robotics & AI) — EIT Digital double degree · [Technische Universität Berlin & KTH Royal Institute of Technology](#) 2018 – 2021

- Full EIT Digital scholarship. Also EIT Digital Summer School, Aalto University, 2019.

BE, Electrical & Electronics Engineering · [Anna University \(Panimalar Engineering College\), Chennai](#) 2013 – 2017

- Partial scholarship. Electives in embedded systems and 3D printing.

PUBLICATIONS AND TALKS

Hoang, M.L. and Shunmugam, N. *ClaimPilot: An Open Multi-Agent Pipeline for Veterinary Insurance Claims Adjudication*. **Future Internet**, 2026, 18(9), 465. doi.org/10.3390/fi18090465 Released a 150-claim open benchmark.

Sreedhar, S. and Shunmugam, N. *Deep Learning for Hardware-Constrained Driverless Cars*. IEEE Conference Publication, 2020. ieeexplore.ieee.org/document/9202838

Shunmugam, N., Jawor, P. and Szyda, J. *Attention-Based Imputation of Structural Variants: genotype prediction in low- and high-linkage-disequilibrium genomes*. Oral presentation (first author), 7th Polish Congress of Genetics, Wroclaw, September 2026. Invited keynote speaker, AI Vet Conference 2024 and 2025. Two manuscripts in preparation (Hirsfeld Institute, 2026).

SKILLS

AI and machine learning: CNNs and Transformers; deep reinforcement and imitation learning; multi-agent systems; LLM fine-tuning and domain specialisation; Python, PyTorch, TensorFlow, scikit-learn, NumPy, pandas; C/C++, MATLAB; familiar with R; Git, Linux, CI/CD.

Genomics and biomedical: CRISPR guide design and off-target analysis; NGS and multi-omics pipelines; k-mer and one-hot sequence encodings; medical and veterinary computer vision; DNA and RNA isolation, PCR, gel electrophoresis, ELISA, cell culture, sequencing library preparation; necropsy, histology and tissue sampling.

Autonomous and embedded systems: object detection and segmentation, depth estimation, LiDAR point clouds, sensor fusion (GPS, IMU, radar), NVIDIA Jetson deployment, PID control, Arduino and IoT, 3D printing and prototyping.

HONOURS AND LANGUAGES

Full EIT Digital scholarship for the TU Berlin / KTH double-degree MSc; partial scholarship for the BE. Entrepreneur First cohort, Berlin 2021. Invited keynote speaker, AI Vet Conference 2024 and 2025. Languages: English (full professional, IELTS C1), Tamil (native), Polish and German (basic). Lived and worked across six countries.